

C.1 Simple harmonic motion

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **29 marks**

Take $g = 9.81 \text{ m s}^{-2}$ unless told otherwise. Show your working; the method earns marks even when the arithmetic wobbles.

Question 1 — The hypnotist's pendulum (9 marks)

A stage hypnotist swings a crystal on a long chain. The crystal, of mass 60 g, hangs from a chain of length 2.5 m and swings through a small angle.

- (a) State the two conditions that define simple harmonic motion. [2]
- (b) Calculate the period of the crystal's swing. [2]
- (c) For dramatic effect, the hypnotist swaps the crystal for one of mass 120 g on the same chain. She claims the audience will now blink twice as slowly. State what actually happens to the period, and justify your answer. [2]
- (d) The act moves to a (hypothetical) theatre on Mars, where $g = 3.7 \text{ m s}^{-2}$. Calculate the new period of the 2.5 m pendulum, and state whether period and gravitational field strength are directly or inversely related. [3]

Question 2 — The bungee trampoline (9 marks)

At the school fair, a 48 kg student bounces on a bungee trampoline. Near the lowest points her motion is simple harmonic, governed by an effective spring constant $k = 150 \text{ N m}^{-1}$, with amplitude 0.40 m.

- (a) Calculate the total energy of her oscillation. [2]
- (b) Calculate her maximum speed. [3]
- (c) A friend claims that doubling the amplitude would double the total energy. Show quantitatively that the friend is wrong. [2]
- (d) State where in the cycle her speed is greatest and where the magnitude of her acceleration is greatest. [2]

Question 3 — [HL] The bounce, written down (5 marks)

For the student in Question 2, take $t = 0$ at the moment she passes through the equilibrium position moving upwards.

- (a) Write an equation for her displacement $x(t)$, with numerical values for the amplitude and the angular frequency. [2]
- (b) Calculate her speed when her displacement is 0.20 m. [3]

Question 4 — The over-oiled grandfather clock

(6 marks)

An antique dealer oils a grandfather clock so enthusiastically that thick oil reaches the pendulum bearing, lightly damping the swing.

- (a) Describe what light damping does to the amplitude of the pendulum over many cycles. [2]
- (b) The dealer says: “at least the period is unchanged, so the clock still keeps perfect time.” Evaluate this claim. [2]
- (c) The dealer’s workshop door has a closer that returns it to the frame as quickly as possible without oscillating. Name this type of damping and state one other application of it. [2]

Answers

Question 1 — The hypnotist's pendulum

- (a) The acceleration is (i) proportional to the displacement from equilibrium and (ii) always directed *towards* equilibrium. Together: $a = -\omega^2 x$.
- (b) $T = 2\pi\sqrt{L/g} = 2\pi\sqrt{2.5/9.81} = 3.2$ s.
- (c) The period is *unchanged*. $T = 2\pi\sqrt{L/g}$ contains no mass term, so doubling the crystal's mass does nothing. (The restoring force grows in proportion to the mass, so the acceleration is unaffected.)
- (d) $T = 2\pi\sqrt{2.5/3.7} = 5.2$ s. Period and g are *inversely* related, though not in direct proportion, since $T \propto 1/\sqrt{g}$: weaker gravity gives a slower swing.

Question 2 — The bungee trampoline

- (a) $E = \frac{1}{2}kA^2 = \frac{1}{2}(150)(0.40)^2 = 12$ J.
- (b) At maximum speed all the energy is kinetic: $\frac{1}{2}mv^2 = 12 \Rightarrow v = \sqrt{2(12)/48} = 0.71$ m s⁻¹. (Equivalently $v_{\max} = A\omega = A\sqrt{k/m}$.)
- (c) Energy goes as the *square* of the amplitude. Doubling to 0.80 m gives $\frac{1}{2}(150)(0.80)^2 = 48$ J, which is 4×12 , not $2 \times 12 = 24$ J.
- (d) Her speed is greatest as she passes through the equilibrium position, where the displacement is zero. The magnitude of her acceleration is greatest at the two extremes of the motion, where the displacement is largest, since $a = -\omega^2 x$.

Question 3 — [HL] The bounce, written down

- (a) $\omega = \sqrt{k/m} = \sqrt{150/48} = 1.77$ rad s⁻¹. Starting at equilibrium and moving upwards makes this a sine, so $x = 0.40 \sin(1.77t)$ metres.
- (b) $v = \omega\sqrt{A^2 - x^2} = 1.77\sqrt{0.16 - 0.04} = 0.61$ m s⁻¹.

Question 4 — The over-oiled grandfather clock

- (a) The amplitude decreases gradually, losing roughly the same *fraction* each cycle, so the decay is exponential rather than linear. Many oscillations pass before the swing dies away.
- (b) Nearly true, but not exactly. Damping slightly *lowers* the frequency of the oscillation, so the period is slightly longer and the clock runs slow. The dealer is also wrong to think a shrinking amplitude is harmless: once the swing is too small, the escapement may not advance at all.
- (c) Critical damping: the system returns to equilibrium in the shortest time without overshooting. Other applications include vehicle suspension, analogue meter needles and building earthquake dampers.